Here p_0 is ground-state electron momentum. Substitution gives

$$E^* = -\frac{2(ke^2)^2mc^2}{(\hbar c)^2} \approx -54.4\ \text{eV} . \qquad (3)$$

Ignoring electron interaction, the helium ground energy E_0 is 2 times this value:

$$E_0 = 2E^* \approx -108.8\ \text{eV} . \qquad (4)$$

The ion energy E^* minus atomic ground energy E_0 is the first ionization energy:

$$E^+ = E^* - E_0 = -E^* \approx 54.4\ \text{eV} . \qquad (5)$$

Method 2: the helium-ion energy is

$$E^* = \frac{p^2}{2m} - \frac{2ke^2}{r} .$$

Insert the ground angular momentum $rp = \hbar$ to rewrite (3):

$$E^* = \frac{\hbar^2}{2mr^2} - \frac{2ke^2}{r} = \frac{\hbar^2}{2m}\left(\frac{1}{r} - \frac{2ke^2m}{\hbar^2}\right)^2 - \frac{2k^2e^4m}{\hbar^2} .$$

Minimum ground energy makes the first term of (4) zero. Thus radius and energy

$$r_0 = \frac{\hbar^2}{2ke^2m} , \quad E^* = -\frac{2k^2e^4m}{\hbar^2}$$

are respectively as in (1), (2).

(II) Here are two methods for helium ground energy E_0 and radius r_0.

Method 1: the mechanical equation

$$\frac{mv^2}{r} = \frac{2ke^2}{r^2} - \frac{ke^2}{(2r)^2} = \frac{7ke^2}{4r^2}$$

and ground-state quantization $rmv = \hbar$ give the radius

$$r_0 = 4\hbar^2 / 7ke^2m , \qquad (6)$$

Thus the helium ground energy is

$$E_0 = 2\left(\frac{p_0^2}{2m} - \frac{2ke^2}{r_0}\right) + \frac{ke^2}{2r_0} = -\frac{49k^2e^4m}{16\hbar^2} ; \qquad (7)$$

Numerical substitution gives

$$E_0 = -\frac{49(ke^2)^2mc^2}{16(\hbar c)^2} \approx -83.4\ \text{eV} , \qquad (8)$$

$$r_0 = \frac{4(\hbar c)^2}{7ke^2mc^2} \approx 0.0302\ \text{nm} .$$

Therefore the helium first ionization energy is

$$E^+ = E^* - E_0 \approx 29.0\ \text{eV} . \qquad (9)$$

This still exceeds the measured helium ionization energy 24.6 eV by 4.4 eV .